

3rd Grade Human Body

(Scheduled Event - 30 minutes)



Description: Participants will be assessed on their understanding of the structure, function, and organization of major human body systems and the organs that comprise them.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring:

- Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

(Optional) Teams may bring:

- Four-function calculator

Safety Requirements:

- None

The Competition: This event may be administered as a written test or as a series of stations which may include diagrams, models, specimens, medical images, data analysis, and problems for students to solve. Content topics will include but are not limited to:

1. Human Body Systems

- Describe the levels of organization within the human body, including cells, tissues, organs, and organ systems.
- Explain how organ systems work together to maintain homeostasis.
- For each body system, identify the major organs and structures, their locations, and their functions. Systems studied will include: Skeletal, Muscular, Digestive, Respiratory, Circulatory, Excretory, Nervous, Sensory, and Endocrine

2. Data Analysis and Applications

- Interpret diagrams, models, charts, tables, medical images, and other representations of human body systems.
- Apply knowledge of anatomy and physiology to identify organs, determine functions, and solve scenario-based problems.

Examples:

- Given a diagram of the human body, identify the major organs in the digestive, respiratory, circulatory, skeletal, muscular, and nervous systems.
- Match organs to their primary functions (e.g., heart pumps blood, lungs exchange oxygen and carbon dioxide).
- Interpret simple diagrams showing how body systems work together (e.g., lungs and heart, muscles and bones).
- Identify bones, joints, or major muscle groups from labeled diagrams or models.

- Use a simple food digestion diagram to trace the path food takes through the digestive system.
- Interpret a heart rate, breathing rate, or body temperature chart before and after exercise.
- Compare healthy and unhealthy X-rays or other simplified medical images (e.g., broken vs. unbroken bone, healthy vs. cavity-affected tooth).
- Determine which body system is responsible for a specific function or action.
- Apply knowledge of body systems to solve everyday scenarios (e.g., "Which system helps you catch a ball?" or "Which organ works harder during exercise?").
- Analyze simple tables or diagrams to determine how exercise, nutrition, hydration, or rest affects the body.

Scoring:

- Points will be awarded for each correct response.

Tiebreakers:

- Ties will be broken by the accuracy or quality of answers to select questions chosen by the event leader prior to the competition.

Resources:

- [Human Body](#)
- [Biology for Kids](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles